$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

$$= 20 + \frac{7.8 - 5}{4} \times 10$$

$$\text{Median} = 27 \text{ years}$$

Example 5.19

The following is the marks obtained by 140 students in a college. Find the median marks

Marks	Number of students
10-19	7
20-29	15
30-39	18
40-49	25
50-59	30
60-69	20
70-79	16
80-89	7
90-99	2



NOTE

In this problem the class intervals are given in inclusive type, convert them into exclusive type and call it as true class interval.

Solution:

Class boundaries	*f*	*Cf*
9.5 -19.5	7	7
19.5-29.5	15	22
29.5- 39.5	18	40
39.5-49.5	25	65
49.5-59.5	30	95
59.5-69.5	20	115
69.5-79.5	16	131
79.5-89.5	7	138
89.5-99.5	2	140
Total	**N =140**	

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